Dr. Peter Fasogbon

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Research Interests: Computer Vision and 3D Geometry; Multi-view Calibration; Industrial 3D Measurement and Metrology; Camera–Laser Systems and Active Vision; Real-Optics Simulation, Lighting, and Distortion Modelling; Real-time Volumetric Systems; Classical and Neural Rendering; Gaussian Splatting; XR (AR/MR/VR); Visual AI Systems.

Work Experience

April 2017 – Present

Senior Researcher
Previously Senior Scientist, Computer Vision
Nokia Technologies, Finland

- Joined Nokia as Senior Scientist, Computer Vision; the subsequent transition to the broader Senior Researcher title in 2025 reflected an expanded remit across Nokia technologies. Throughout, I have remained the principal computer-vision expert within a multidisciplinary team of compression, streaming, media-systems, and standards specialists.
- Own the technical direction of computer-vision workstreams and lead R&D across multi-camera calibration, depth processing, 3D reconstruction, neural rendering, Gaussian Splatting, volumetric media, and XR systems.
- Collaborate with Nokia Bell Labs, product groups, and specialists in compression, streaming, media systems, cloud platforms, and standardisation to integrate computer vision methods into end-to-end systems, resulting in patents, publications, standards contributions, open-source components, executive demonstrations, and product-oriented initiatives.
- Provide technical direction to external software engineers and cloud specialists, and production partners for computer vision related implementations.
- Supervise and mentor university-affiliated interns on one to two year placements, including Tampere, Aalto University students, by defining objectives, guiding technical work, reviewing results, and integrating successful outcomes into Nokia projects.

April 2013 – April 2016

R & D Engineer, Computer Vision Systems
French Railway Company (SNCF)

- Developed industrial-grade 3D vision systems for monitoring high-speed railway catenary and contact wires.
- Designed calibration and reconstruction pipelines for camera and laser-based inspection systems operating under extreme conditions.
- Performed system testing, validation, and field integration in real operational railway environments.
- Collaborated with academic and industrial partners to transition research prototypes into deployable inspection solutions.

November 2012 – March 2013

Engineer, Computer Vision Simulation
Université de Lille, CRISTAL (CNRS)

- Developed a 3D vision simulation framework for railway inspection applications.
- Modelled camera and laser sensor placement, optical distortions, and environmental perturbations under high-speed railway constraints.
- Implemented simulation components for laser-stripe sensors, stereo vision, refractive distortion, and realistic noise models.

January 2012 – June 2012

Master Research Intern, Computer Vision
Université d'Auvergne, ALCOV ISIT (CNRS)

- Conducted research on real-time tool and tissue segmentation for minimally invasive surgery.
- Developed monocular 3D reconstruction methods for surgical scenes.
- Implemented GPU-accelerated algorithms using CUDA to meet real-time performance constraints.

May 2011 – September 2011

Summer Research Intern, Computer Vision
Université de Bourgogne, Le2i (CNRS), France

- Developed probabilistic image-analysis methods for industrial tube-crack detection.
- Designed statistical correlated filters under exponential noise models.
- Project conducted in collaboration with a multinational manufacturer of tubes for the oil and gas industry.

Education

Sept. 2013 – Oct. 2016

Doctor of Philosophy (Computer Vision)

Université de Lille, France

- Industrial PhD conducted in collaboration with SNCF (French National Railway Company) and Université de Lille, CRISTAL (CNRS).
- Research focused on 3D vision-based dimensional measurement for industrial inspection systems.
- Dissertation: *Dimensional Measurement of Metallic Objects by 3D Vision* (industrial thesis, confidential for four years).

2010 – 2012

Master of Science - VIBOT

Université de Bourgogne, Europe, France

- Erasmus Mundus international master's programme in computer vision and robotics.
- Triple-degree programme with Heriot-Watt University (UK) and Universitat de Girona (Spain).
- Strong focus on multi-view geometry, robotics, image processing, and 3D reconstruction.

2009 – 2010

Professional Bachelor's Degree

Université Joseph Fourier, IUT1, France

- Major in Computer Networks and Telecommunications.
- Final project: Computer Network Security, including firewall design and deployment.
- Technical training in database-backed web systems using MySQL.

2007 – 2009

Undergraduate - Electronics Engineering

Obafemi Awolowo University, Nigeria

- Completed two years toward a B.Eng. in Electronics Engineering.
- Memoir: ZigBee wireless networking, submitted to Université Joseph Fourier.

Project Experience

Nokia Gaussia (2025 – Present)

• Own the computer-vision research direction for next-generation Gaussian Splatting systems for real-time XR within a multidisciplinary team covering compression, streaming, media systems, and standards.

• Research spans capture, representation, rendering, hybrid splat–mesh methods, camera-agnostic pruning, metadata signalling, learned 3D representations, compression, and real-time streaming, in collaboration with Nokia Bell Labs and business units.

• Supervise university-affiliated interns working under the project, defining objectives, guiding experiments, reviewing results, and integrating successful research into the workstream.

Nokia VolStream (2020 – 2025)

• Served as the principal computer-vision expert and led the computer-vision workstream within a multidisciplinary team of video-compression, streaming, media-systems, networking, and standards specialists.

• Designed multi-view calibration, depth refinement, 3D reconstruction, and classical and neural rendering pipelines integrated with V-PCC and MIV compression technologies; contributed to OMAF and HEIF camera-parameter signalling.

• Provided technical direction to external software and cloud engineers during integration and deployment, and coordinated capture setup, calibration, and validation with external field and production partners.

*• Outcomes include patents, publications, standards contributions, demonstrations, and the **Nokia VolStream open-source repository**. The full multi-camera calibration implementation is under review for a future public release.*

Impact & Public Demonstrations:

Mobile World Congress (MWC): *2023, 2024, 2025 - Live demonstrations of VolStream, including Cloud RAN integration with Nokia network infrastructure.*

Brooklyn 6G Summit: *2025 - Joint demonstration of VolStream with IVAS (Immersive Voice and Audio Services), enabling live spatial audio transmission over 5G networks.*

IBC Amsterdam: *2025 - XR demonstrations using lightweight smart glasses (XREAL Air 2), with support for widely used RGB-D sensors such as Azure Kinect and Orbbec cameras.*

GStreamer Conference: *2024 - End-to-end real-time volumetric streaming demonstrations and integration with GStreamer-based media pipelines.*

Millennium Technology Prize Forum: *2024 - Public demonstration of real-time volumetric communication technologies.*

Nokia Radio World: *2023 - Hosted live demonstrations showcasing volumetric media over Nokia radio and network technologies.*

Executive Demonstrations: *2024 - Private demonstration to Nokia CEO, executive leadership, and Nokia Technologies leadership teams.*

Nokia MD3C (2019 – 2020)

• Owned and led the MD3C computer-vision workstream end to end within a multidisciplinary team, including research direction, architecture, method selection, development, validation, and demonstrations.

• Developed a mobile depth-generation system combining smartphone computer vision and cloud-assisted processing, while coordinating technical interfaces with software, cloud, media, and platform specialists.

*• The work supported a winning internal pitch that led to the creation of the **Nokia RXRM** (Real-time eXtended Reality Multimedia) initiative.*

• Demonstrated the system to Nokia Technologies leadership and its President; additional outcomes included early concepts for 3D-aware overlays in standard OMAF players.

CAMESCAT (2013 – 2016)
*• Industrial PhD project conducted within the CAMESCAT programme, led by SNCF Ingénierie and supported by regional, national, and European funding (**SNCF overview**).*
• Developed advanced vision-based systems for direct, real-time measurement and inspection of high-speed railway catenary wires, enabling scalable industrial deployment and reducing reliance on foreign technologies.
*• Total project budget of approximately **3.3 million EUR** over **36 months**, with inter-ministry regional funding, partial EU support, and additional contributions from local authorities (**Optitec project summary**).*
• Main scientific contributor and technical liaison during my industrial PhD, bridging academic research and industrial deployment across an international consortium.
• Technical work covered camera and laser calibration, real-time 3D reconstruction, optical and distortion modelling, non-conventional sensor configurations, and active system placement for high-speed environments.

Industrial & Academic Collaboration:

- ***SNCF Ingénierie*** *(Project lead, France)*
- ***CSEM - Swiss Center for Electronics and Microtechnology*** *(csem.ch) - optics, lighting, and sensor simulation*
- ***MERMEC Group*** *(mermecgroup.com) - collaboration across France, Australia, Japan, and Canada*
- ***Ateliers Laumonier*** *(laumonier.fr)*
- ***O2GAME*** *(o2game.com)*
- ***Université de Lille / CRISTAL (CNRS)***

• Delivered robust prototypes validated in real railway environments, with strong media visibility, patent-oriented outcomes, and foundations for industrial commercialisation and technology transfer to metrology and quality control.

Other Work Experience

Transcriber at Systrad (2013)
Part-time: Translation tasks for the French National Police in Lille, France (English–French).

Publications and Patents

Published 15+ peer-reviewed articles as first author and co-author in computer vision and engineering, with over 90 citations.

Author or co-author of 5+ publicly available patents, with more than 10 additional patent applications across computer vision, volumetric media, immersive systems, compression-related technologies, and collaborative Nokia research.

Full list available on **Google Scholar**

Hobbies

Gardening, travelling, football goalkeeping, dancing, and playwriting.

Awards

2025 **Best Paper Nomination (Top 3)**
27th International Symposium on Multimedia (ISM), Naples, Italy - Paper on multi-camera colour calibration for volumetric streaming.

2020 **Nokia Best Pitch Award**
*Winning internal pitch on OneReality that contributed to the launch of **Nokia RXRM** (Real-time eXtended Reality Multimedia).*

2012 **Merit-Based Grant for PhD Thesis**
Inter-Ministry Fund of Nord-Pas-de-Calais Region, France

2010 **Merit-Based Grant - CISCO "More Together" Competition**
IPv6 competition, third place nationwide in France

Languages

English	Fluent / Full Professional Proficiency
French	Full Professional Proficiency
Finnish	Intermediate

Skills

Leadership	Computer-vision domain and workstream ownership; technical coordination across several technological discipline; intern supervision and mentoring
Programming	C/C++, CUDA, Python, MATLAB, Java, Scala
Libraries & Frameworks	OpenCV, ROS, Ceres, g2o, OpenCL, OpenGL, Blender, Unity
Visual AI & ML	PyTorch, TensorFlow, depth processing, neural rendering, Gaussian Splatting, and learned 3D representations
3D Vision & XR	SLAM, SfM, multi-view geometry, structured light, camera–laser and RGB-D systems, volumetric media, mesh and animation models, and real-time processing